CAE 2026

TANDIL / ARG









Reliability-Centered Maintenance: a short introduction to Industrial Engineering through Reliability models and Multivariate Statistical Process Monitoring

Dr. Diego Nascimento



JOINT WORK WITH

CAE 2026
TANDIL / ARG



Dr. Francisco Louzada



Dr. Paulo Henrique



Dra. Lilia Costa



Dr. Pedro Ramos

Universidade de São Paulo

CeMEAI

CER
Centro de Estudos do Risco

UFBA
Universidade Federal da Bahia

AGENDA (3 sections - DAY #1)

CAE 2026
TANDIL / ARG

SOCIEDAD ARGENTINA DE ESTADÍSTICA

Grupo Argentino de Bioestadística

- Addressing Challenges in INDUSTRIAL STATISTICS
  - Univariate Control Charts (parametric v/s non-parametric)
- An Alternative for Multivariate Statistical Process Monitoring (SPM)
  - Hotelling T^2
  - Extensions for Control Charts (Elliptical family)
  - Joint Distribution estimation using Copula
- Symbolic Data Representation (SDA)
  - Dynamic-Symbolic SPM
- Causal TS (Dynamic Linear Model + Causal Inference)
  - Non-stationary and non-Gaussian (Inclusion of Volatility Models)

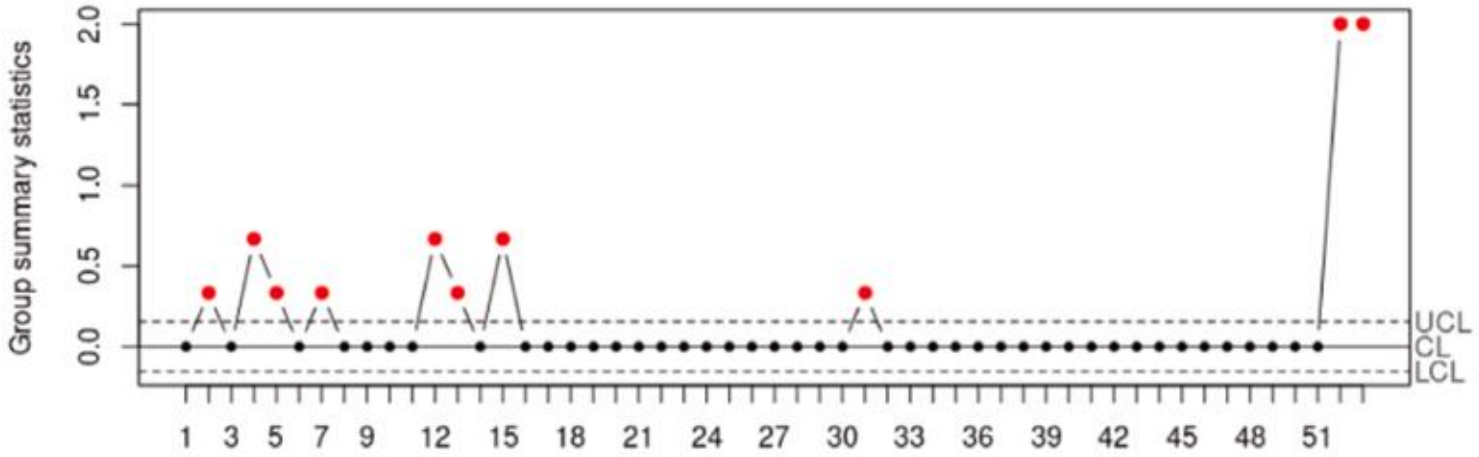
Addressing Challenges in INDUSTRIAL STATISTICS

Industry 4.0 changed the dynamic of generating data and analyzing to support the decision-making. Alongside comes challenges like,

- Multisensory, High-frequency, Spatiotemporal dependence

Colosimo, Bianca M., et al. "Statistical process monitoring from industry 2.0 to industry 4.0: Insights into research and practice." *Technometrics* 66.4 (2024): 507-530.

These sensors enable the connection from the physical to the digital world, providing real-time information on everything from machine performance to product quality (and new data types to be analyzed).



[Session 1] UNIVARIATE SPC

Univariate (Parametrics and Nonparametrics)

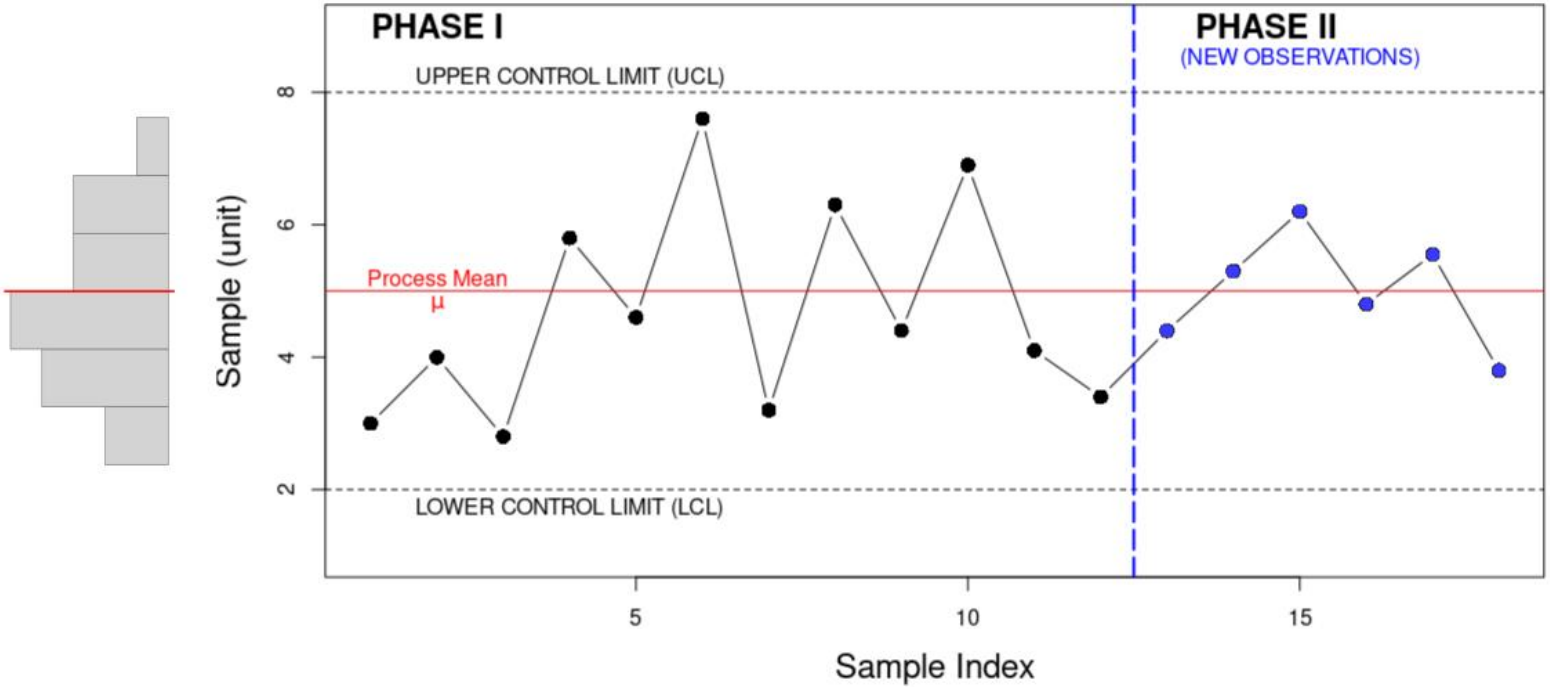
Addressing Challenges in INDUSTRIAL STATISTICS

Statistical Process Monitoring (SPM) is a powerful set of tools used to solve (visualize) problems in order to explain variability and plan/achieve stability of productive performance or services rendered (Montgomery, 2013). For instance,



- The $\bar{X}$ (mean or average) control chart, adopted for process monitoring and control in the **univariate case** (Walter SHEWHART in 1924).
- The Hotelling's T^2 control chart (Hotelling, 1947), which is a **multivariate approach** that considers jointly varying variables and assumes that they are multivariate normally distributed and serially uncorrelated.

Univariate Shewhart's Control Chart (Time-Invariant Parameters)



HISTORY CONTROL CHART

Shewhart's 1924, the Western Electric Handbook, and the founding of American Society for Quality Control (ASQC) and the Journal of Quality Technology, requiring statistical quality control in wartime production.

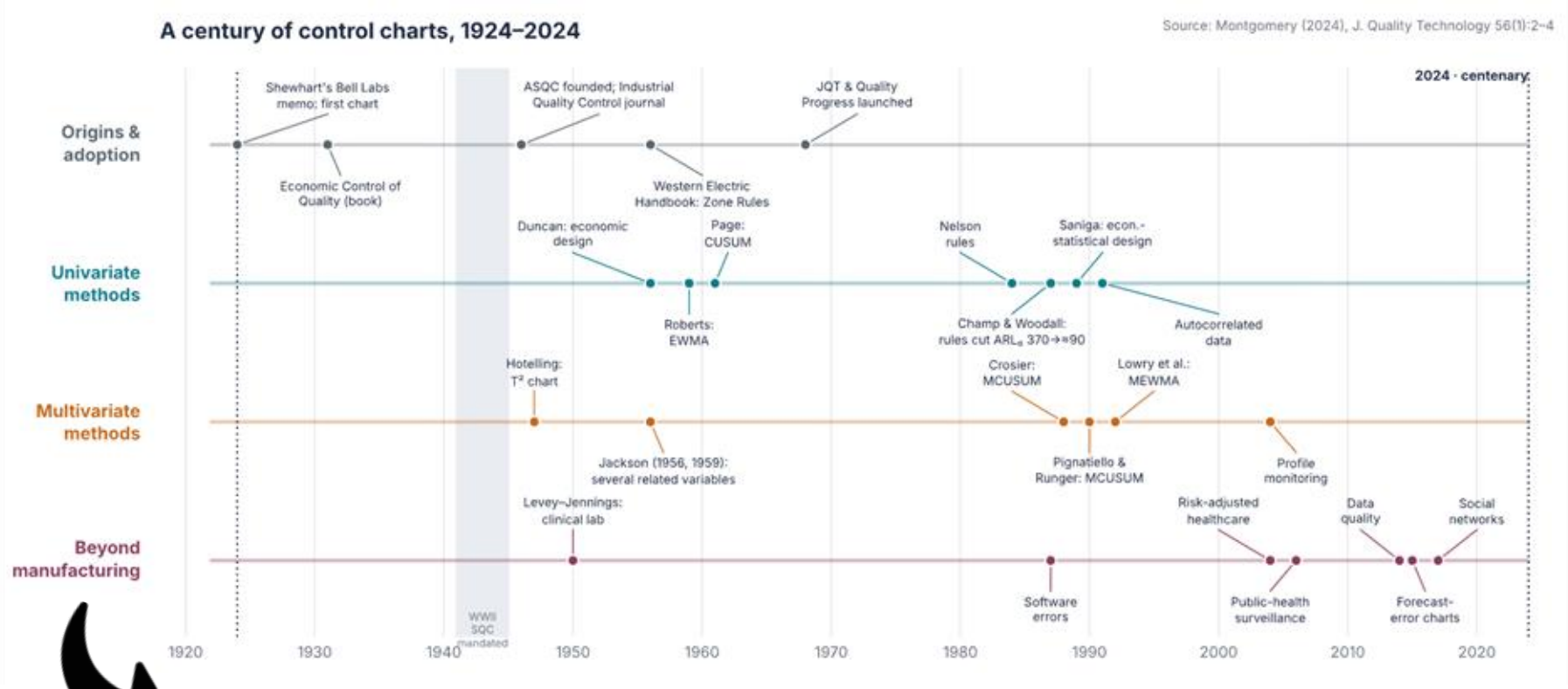
- **Univariate methods:** Shewhart (x-bar, s-chart and others), CUSUM, EWMA, the Nelson rules, economic design, and charts for autocorrelated data.
- **Multivariate methods:** Hotelling T^2, MCUSUM, MEWMA, profile monitoring and others.

Shewhart's core idea: separate chance causes from assignable causes; 3σ limits give an in-control average run length (ARL_0) of about 370 samples.

- **Sensitivity vs. false alarms:** the Zone Rules catch small shifts but cut ARL_0 to about 90; CUSUM and EWMA catch small shifts without the extra false alarms.
- **Univariate to multivariate:** T^2 → MCUSUM → MEWMA → profiles.

Beyond manufacturing: clinical labs, healthcare, public-health surveillance, software, data quality, forecast errors, and social networks. Extensions adopting Machine Learning methods exist.

MONTGOMERY, Douglas C. The 100th anniversary of the control chart. **Journal of Quality Technology**, v. 56, n. 1, p. 2-4, 2024.



FURTHER READING: Boaventura, L. L., Ferreira, P. H., & Fiaccone, R. L. (2022). On flexible statistical process control with artificial intelligence: classification control charts. *Expert Systems with Applications*, *194*, 116492.

UNIVARIATE CONTROL CHARTS

CAE 2026
TANDIL / ARG

Contents lists available at ScienceDirect

Array

journal homepage: www.elsevier.com/locate/array

Nonparametric SPC: Solving complex industrial monitoring tasks with low computational cost

Diego C. Nascimento [a], Anderson O. Fonsêca [b], Marcelo B. Fonsêca [b], Moisés A.F. Silva [b], Paulo H. Ferreira [b,*]

[a] NEOMA Business School, 76825 Mont-Saint-Aignan, France
[b] Department of Statistics, Institute of Mathematics and Statistics, Federal University of Bahia, Salvador, 40170110, Bahia, Brazil

ARTICLE INFO

Dataset link: https://github.com/CER-UFBA/npSPC

Keywords:
Control chart
Distribution-free
Industrial statistics
Nonparametric statistics
Real-time monitoring

ABSTRACT

Quality assurance is essential for an organization's success today, but it is also a great challenge in the Big Data era. Statistical Process Control (SPC), focusing on control charts of monitoring variability, is a fundamental tool, though outliers, skewed information, and temporal dependence may influence the ones based on the process average. Nonparametric techniques stand out when it is needed to represent the process dynamism with robustness and derive statistical tests based on, e.g., the median instead of the mean. This work approaches the univariate Nonparametric Process Monitoring, such as Sign and Signed-Rank statistics, which are useful for run-type signaling, monitoring spread or joint (considering the known parameter of the specified tolerant interval, case K, or unknown parameter of in-control, case U). This study illustrates some mining industry monitoring tasks through nonparametric control charts, allowing the suggestion of a preventive maintenance

SUMMARY NONPARAMETRIC CONTROL CHARTS

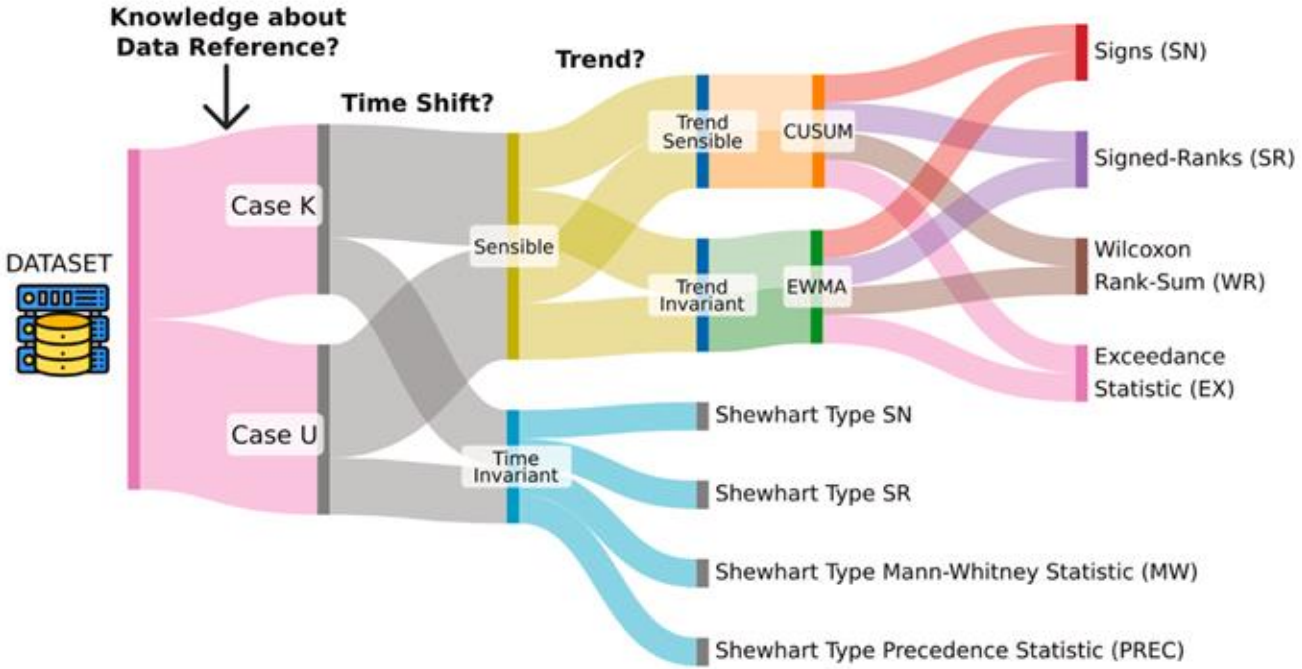
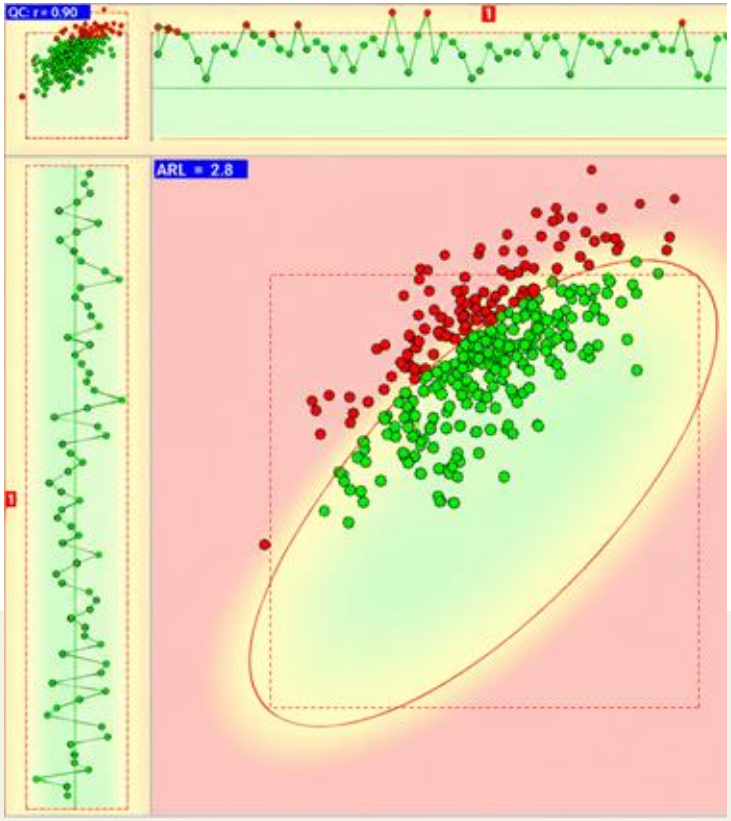


Fig. 1. Sankey diagram describing the working flow of the npSPC package and its implemented class of functions.

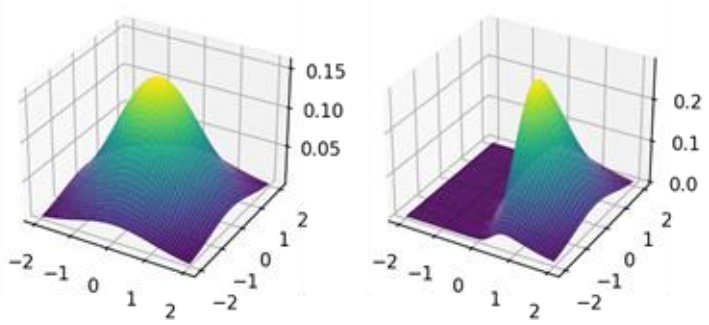


[Session 2] MULTIVARIATE SPC

Full Symmetric (or generalize normality) v/s Univariate Models + Copula

SPC multivariate (Hierarchical + Modified Dist.)

Extensions of a univariate Shewhart control chart were also seen through the Hotelling's T^2 control chart (Hotelling, 1947), which is a multivariate approach that considers jointly varying variables and assumes that they are multivariate normally distributed (with and without serially uncorrelated errors).



Modified-normal distributions, independent & identically distributed, can be seen as a multivariate modified-normal distribution (with a typically requires specific relationships between the mean vector and covariance matrix). For instance, the Skew Normal distribution, Alpha-Skew Normal, and others Generalized Normal distributions can extend the Hotelling's T^2 multivariate control chart concept (Elliptical/circular Regions of Acceptance).

T2 de Hotelling



Motivation: why not p separate univariate charts?

- **False alarms add up.** With p independent charts, each with false-alarm rate α, the overall rate is

  $1 - (1 - \alpha)^p$. For p = 4 at 3σ that is about 1%, not 0.27%.

- **Correlation is ignored.** p separate charts give a *rectangular* in-control region. The true region for correlated normal data is an *ellipse*. So some points look fine on every individual chart but are clearly unusual jointly, and some flagged points are actually normal.

The statistic - T^2 is a squared Mahalanobis distance of a new observation (or subgroup mean) from the in-control centre, scaled by the covariance:

- **Subgroups of size n:** $T^2 = n\,(\bar{x} - \bar{\bar{x}})^T\, S^{-1}\, (\bar{x} - \bar{\bar{x}})$
- **Individual observations:** $T^2 = (x - \bar{x})^T\, S^{-1}\, (x - \bar{x})$

When μ and Σ are known, $T^2 \sim \chi^2_p$, and the upper limit is simply $\chi^2_{p,\alpha}$. There is only an upper limit, since $T^2 \geq 0$.

SPC multivariate (Flexible Marginals + COPULA)

A control chart can be seen as a probabilistic function (Parametric, Semi- or Nonparametric) with regression component (Mandel, 1969), to monitor a statistical process measurement (such as its rolling-window location, dispersion, etc).

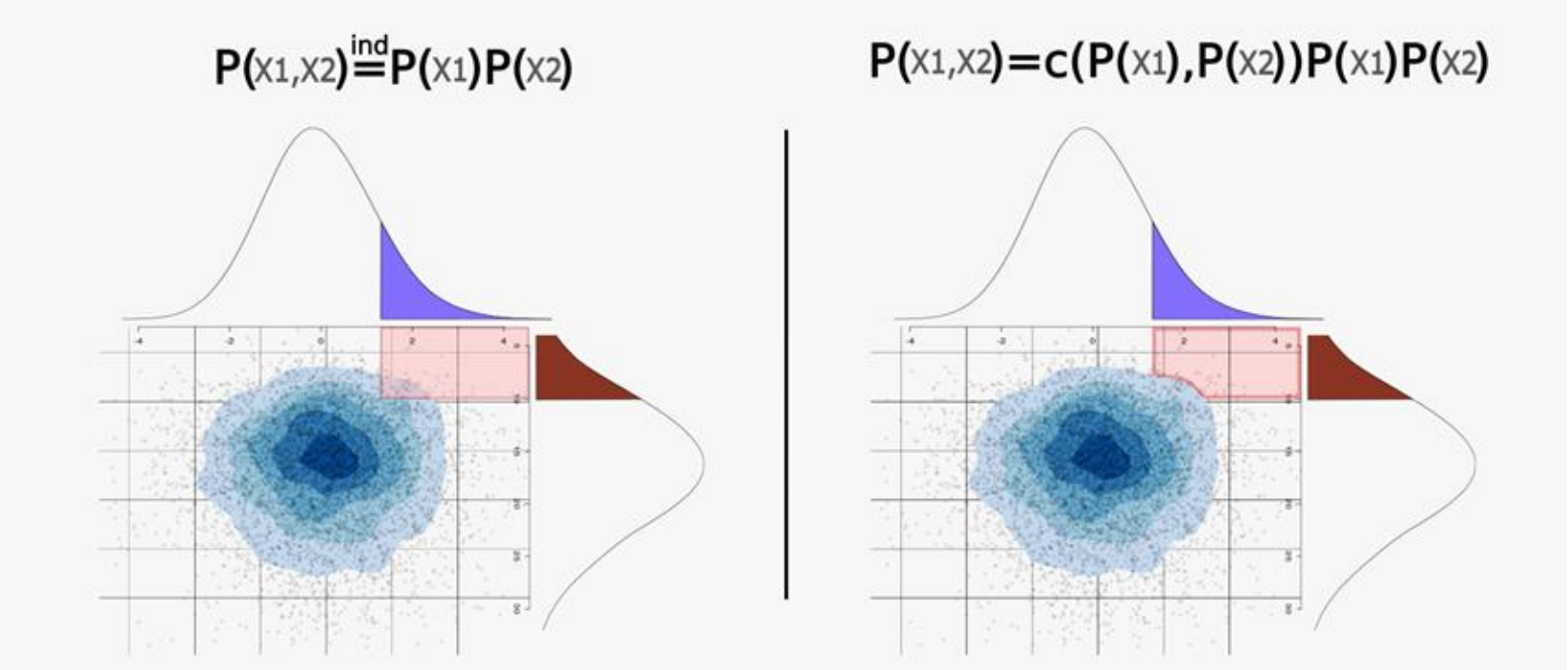
Then, univariate marginal distributions, associated with confidence regions, can be wrapped up with Copula theory (Sklar's theorem; Sklar, 1959) to draw the joint distribution, h(.), relationship across the multivariate response, (X_1, X_2, …, X_d).

$$h(X_1, X_2, \ldots, X_d) = c(F_1(X_1), F_2(X_2), \ldots, F_d(X_d)) \prod_{j=1}^{d} f_j(X_j).$$



Copula	$C(u,v)$	$c(u,v)$	$\theta \in$
Gaussian	$\int_{-\infty}^{\Phi^{-1}(u)} \int_{-\infty}^{\Phi^{-1}(v)} \frac{1}{2\pi(1-\theta^2)^{\frac{1}{2}}} \times \exp\left\{-\frac{s^2-2\theta st+t^2}{2(1-\theta^2)}\right\} dtds$	$\sqrt{\frac{1}{1-\theta^2}} \exp\left\{-\frac{\zeta_1^2-2\zeta_1\zeta_2+\zeta_2^2}{2(1-\theta^2)}\right\} \exp\left\{\frac{1}{2}(\zeta_1^2+\zeta_2^2)\right\}$ where $\zeta_1 = \Phi^{-1}(u)$ and $\zeta_2 = \Phi^{-1}(v)$	$[-1,1]$
Student's t_ν	$\int_{-\infty}^{T_\nu^{-1}(u)} \int_{-\infty}^{T_\nu^{-1}(v)} \frac{1}{2\pi(1-\theta^2)^{\frac{1}{2}}} \times \left(1+\frac{s^2-2\theta st+t^2}{\nu(1-\theta^2)}\right)^{-\frac{\nu+2}{2}} dtds$	$\frac{1}{2}\left(\frac{\Gamma(\frac{\nu}{2})}{\Gamma(\frac{\nu+1}{2})}\right)\nu(1-\theta^2)^{-\frac{1}{2}} \times \left[\left(1+\frac{[T_\nu^{-1}(u)]^2}{\nu}\right)\left(1+\frac{[T_\nu^{-1}(v)]^2}{\nu}\right)\right]^{\frac{\nu+1}{2}} \times \left(1+\frac{[T_\nu^{-1}(u)]^2-2\theta T_\nu^{-1}(u)T_\nu^{-1}(v)+[T_\nu^{-1}(v)]^2}{(1-\theta^2)\nu}\right)^{-1-\frac{\nu}{2}}$	$[-1,1]$
Clayton	$\left(u^{-\theta}+v^{-\theta}-1\right)^{-\frac{1}{\theta}}$	$(1+\theta)(uv)^{-1-\theta}\left(u^{-\theta}+v^{-\theta}-1\right)^{-\frac{1}{\theta}-\theta}$	$(0,\infty)$
Frank	$-\frac{1}{\theta}\log\left(1+\frac{[\exp\{-\theta u\}-1][\exp\{-\theta v\}-1]}{\exp\{-\theta\}-1}\right)$	$\frac{\theta[1-\exp\{-\theta\}]\exp\{-\theta(u+v)\}}{1-\exp\{-\theta\}-(1-\exp\{-\theta u\})-(1-\exp\{-\theta v\})}$	$\mathbb{R}-\{0\}$
Gumbel	$\exp\left\{-\left[(-\log u)^\theta+(-\log v)^\theta\right]^{\frac{1}{\theta}}\right\}$	$\frac{(-\log u)^{\theta-1}(-\log v)^{\theta-1}}{uv} \times \exp\left\{-\left[(-\log u)^\theta+(-\log v)^\theta\right]^{\frac{1}{\theta}}\right\} \times \left(\left[(-\log u)^\theta+(-\log v)^\theta\right]^{\left(\frac{1-\theta}{\theta}\right)^2}\right) + \left([\theta-1]\left[(-\log u)^\theta+(-\log v)^\theta\right]^{\left(\frac{1-2\theta}{\theta}\right)}\right)$	$[1,\infty)$

$$\int_{\{\hat{h}_n(\boldsymbol{x}) \geq \hat{c}_n\}} \hat{h}_n(\boldsymbol{x}) d\boldsymbol{x} = 1 - \alpha.$$

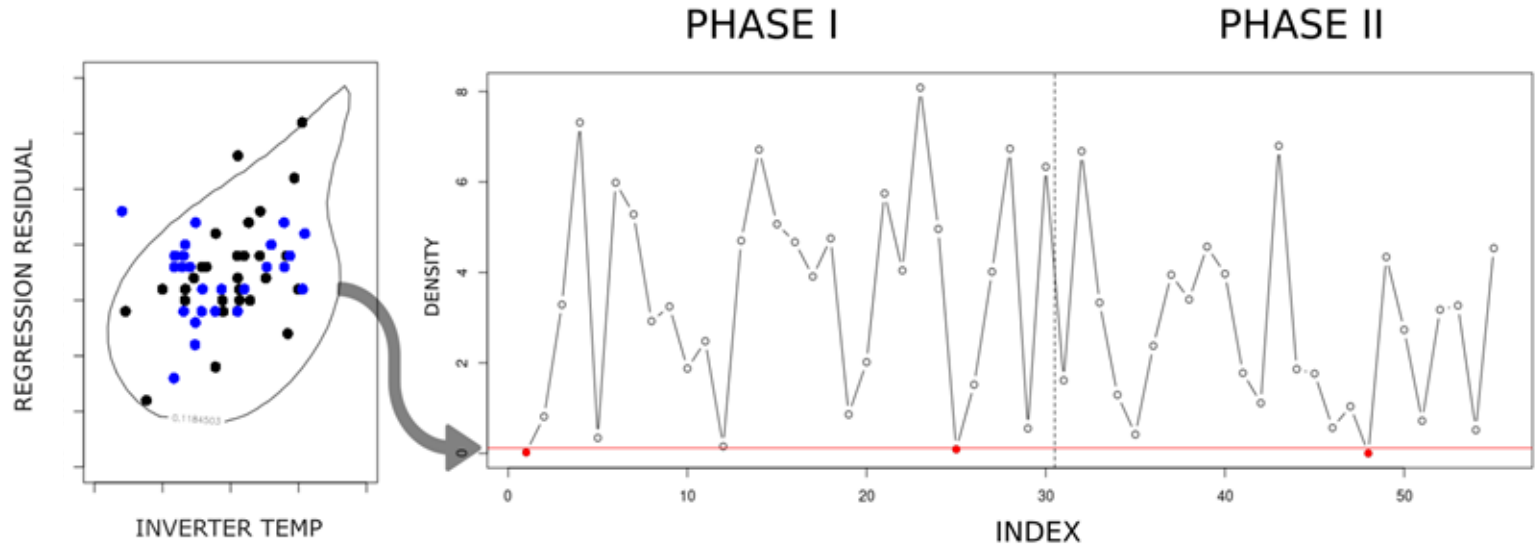


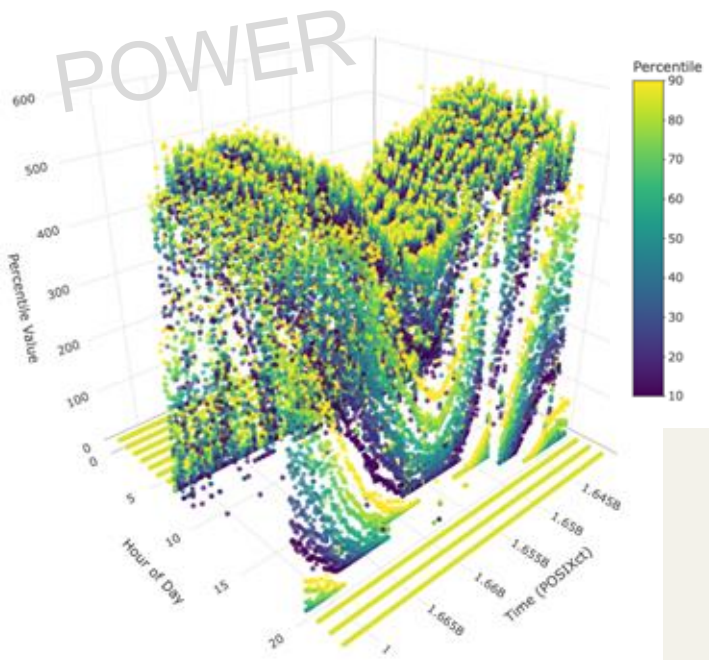
Visual representation of the Region of Rejection, but concept is extendable for dimension p > 2.

SPC multivariate (Flexible Marginals + COPULA)

CAE 2026
TANDIL / ARG

Any time series model for the POWER called regression, then i.e. daily perform represented by the averages of model's estimated error & Inverse Temperature,



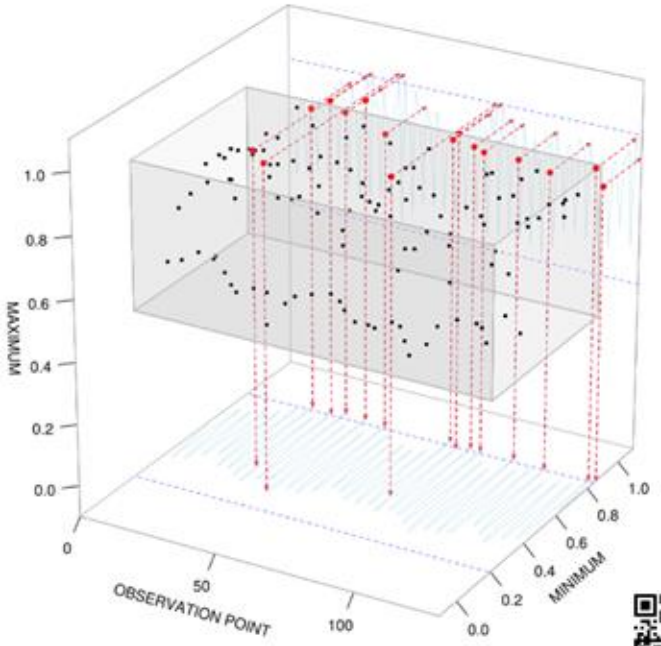


[Session 3] EXTENDING THE SPC

Time Series Symbolic Data Representation

Bivariate Control Chart for Interval Time Series (TS)

CAE 2026
TANDIL / ARG



Previews models focus on the process mean, but SPM is far beyond only centrality measurement, especially if the interest is INDIVIDUAL performance.

An Uniform Interval representation was proposed for TS Symbolic Data Representation (including tolerances in the Lower and Upper limits) as a Bivariate Control Chart tool considering independence among the bounds. Then,

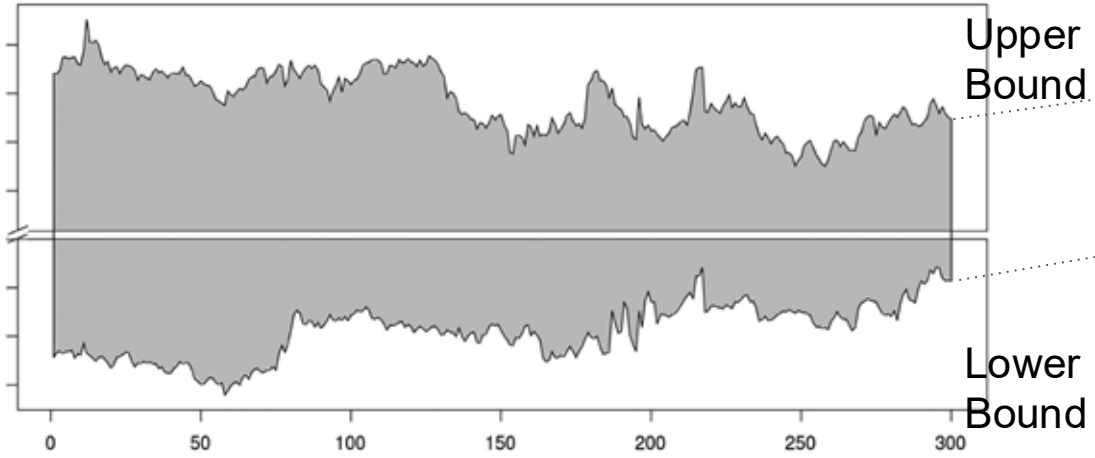
Acceptance Region of the Control Chart is rectangular (and, initially, time-invariant).

Dynamic Symbolic Interval for Time Series

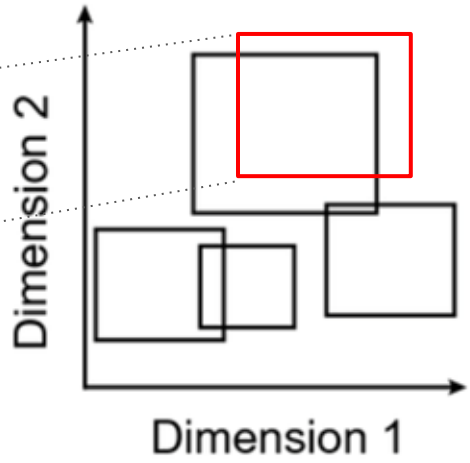
CAE 2026
TANDIL / ARG

REPRESENTATION	OBS. (n)	TOTAL (%)
Univariate SPM	5,202,600	-
Interval SPM	868,550	16.70%

BIVARIATE
REPRESENTATION

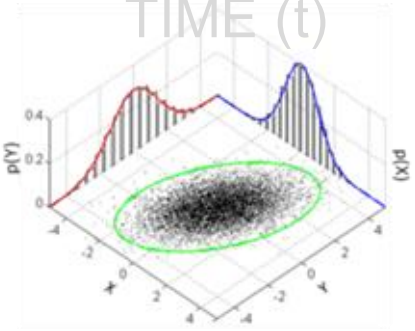


TIME
MARGINAL SDA



Acceptance Regions

*UNIFORM RECTANGULAR ACCEPTANCE REGION



Dynamic Symbolic Time Series

CAE 2026
TANDIL / ARG

(explanation including time-varying variation)

SCAN ME

Dynamic Linear Model
(DLM) framework

$$Yt \mid Yt\text{-}1 \sim N_m(\mathbf{F_t\mu_t}, \mathbf{V_t})$$

[KALMAN FILTER]

$$\mu_0 \sim N_p(\theta_0, \sigma_0^2)$$

after time $t \geq 1$ then,

$$\underbrace{Y_t = F_t\mu_t + \upsilon_t}_{\text{observation equation}}, \quad \upsilon_t \sim N_m(0, V_t),$$

$$\underbrace{\mu_t = G_t\mu_{t-1} + \omega_t}_{\text{state equation}}, \quad \omega_t \sim N_p(0, W_t)$$

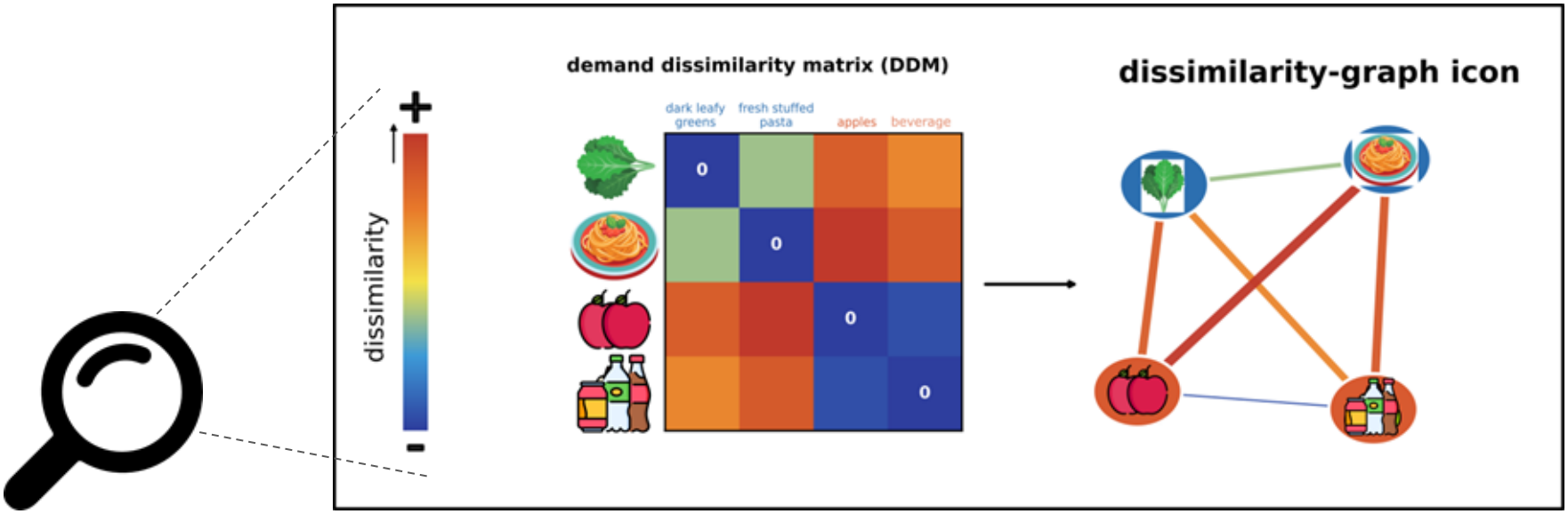


Wrapping-up

CAE 2026
TANDIL / ARG

Usefulness of the presented Control Chart tools for different monitoring levels…

METHODOLOGY	Invariante CI	Time-varying Tolerance	Reduced Dimension	Rectangular Region	Irregular TS
Univariate (Parametric or Nonparametric)	X	X			
Joint Distribution through Copula	X	X			
Ellipse Multivariate Family (T-Student, Normal, and extensions w/ Multivariate Distributions)	X	X			
Symbolic Data Representation (SDA)	X	X	X	X	
Dynamic SDA		X	X		X





[Session 4] TS with Causal Inference

Multiregression Dynamic Model (MDM)

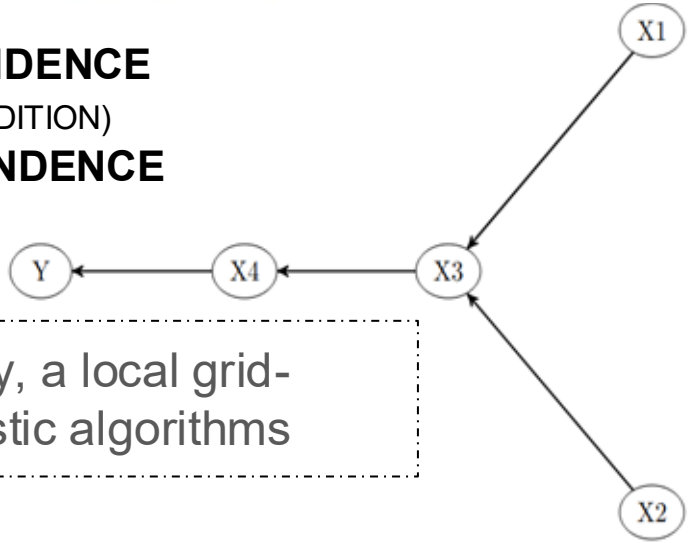
HOW to represent CAUSALITY in a PROCESS? (via DAG)

FACTORIZATION OF JOINT PROBABILITY DENSITY FUNCTION

$$\mathbb{P}(X1, X2, X3, X4, Y) = \mathbb{P}(X1)\mathbb{P}(X2|X1)\mathbb{P}(X3|X2, X1)\mathbb{P}(X4|X3, X2, X1)\mathbb{P}(Y|X4, X3, X2, X1)$$
$$\overset{*}{=} \mathbb{P}(X1)\mathbb{P}(X2)\mathbb{P}(X3|X2, X1)\mathbb{P}(X4|X3)\mathbb{P}(Y|X4)$$

* CONDITIONAL INDEPENDENCE

STATISTICAL INDEPENDENCE
= (if CAUSAL MARKOV CONDITION)
GRAPHICALLY INDEPENDENCE



Complexity is a NP-hard problem! Alternatively, a local grid-search using Bayes Factor may help or Heuristic algorithms

Class of multivariate state space TIME SERIES models,

- Multiregression Dynamic Model (MDM)

It's a class of non-Gaussian time series models which decompose the forecasting system into components whose conditional distributions are univariate Bayesian dynamic linear regression models (DLMs) through Direct Acyclic Graph (DAG) representation.

- Intervention, trends and seasonals, can be transferred directly on to these models with no appeal to stringent symmetry conditions (QUEEN & SMITH, 1993).

Those components are hypothesized to exhibit certain **conditional independence**. For instance, any random vectors **X**, **Y**, **Z** and **W** will have the properties:

$$\begin{aligned}
&\text{(a)} \quad \mathbf{X} \amalg \mathbf{Y} \mid (\mathbf{Y}, \mathbf{Z}), \\
&\text{(b)} \quad \mathbf{X} \amalg \mathbf{Y} \mid \mathbf{Z} \Leftrightarrow \mathbf{Y} \amalg \mathbf{X} \mid \mathbf{Z}, \\
&\text{(c)} \quad \mathbf{X} \amalg (\mathbf{Y}, \mathbf{Z}) \mid \mathbf{W} \Leftrightarrow \begin{cases} \mathbf{X} \amalg \mathbf{Y} \mid (\mathbf{Z}, \mathbf{W}) \\ \text{together with} \\ \quad \mathbf{X} \amalg \mathbf{Z} \mid \mathbf{W} \end{cases}
\end{aligned}$$

where $\mathbf{X} \amalg \mathbf{Y} \mid \mathbf{Z}$ reads **X** is independent of **Y** given **Z** (see Dawid (1979)).

FIRST STEP - NETWORK STRUCTURE ESTIMATION

Costa et al. (2015) propose the MDM-IPA straightforward search over all possible connectivity networks, DAGs, with dynamically changing intensity of transmission to find the Maximum a Posteriori (MAP) probability.

Searching MDM Using IPA

The number of nodes (n)	Sample Size (T)			
	100	200	600	1200
3	0.13	0.25	0.75	1.51
4	0.35	0.68	2.05	4.09
6	3.79	7.39	21.15	39.94
11	167.99	325.50	1001.87	1982.08

Table 1: The time in minutes to find the scores for different numbers of nodes and sample sizes, and the discount factor was chosen in the range from 0.5 to 1.0 with increments of 0.01.

Alternatively, my current searches focus on optimizing this optimal search using Heuristics algorithms or restriction to speed-up the estimation process.

CAE 2026
TANDIL / ARG

Contents lists available at ScienceDirect

Software Impacts

journal homepage: www.journals.elsevier.com/software-impacts

Original software publication

MDM: An R package for causal multivariate time series tasks

Lilia Costa [a], Arthur Azevedo [a], Michel Miler Rocha dos Santos [a], Diego Carvalho Nascimento [b,*]

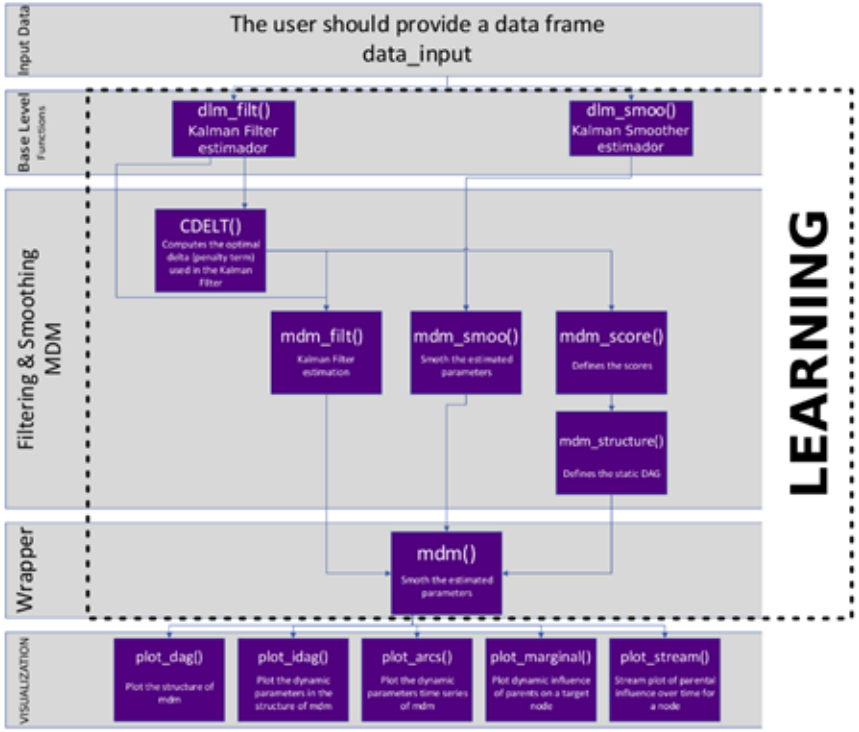
[a] Department of Statistics, Federal University of Bahia, Salvador, Brazil
[b] NEOMA Business School, Mont-Saint-Aignan, 76825, France

ARTICLE INFO

Keywords:
Causal time series
Common-structure estimation
Dynamic coefficients
Non-gaussian multivariate time series

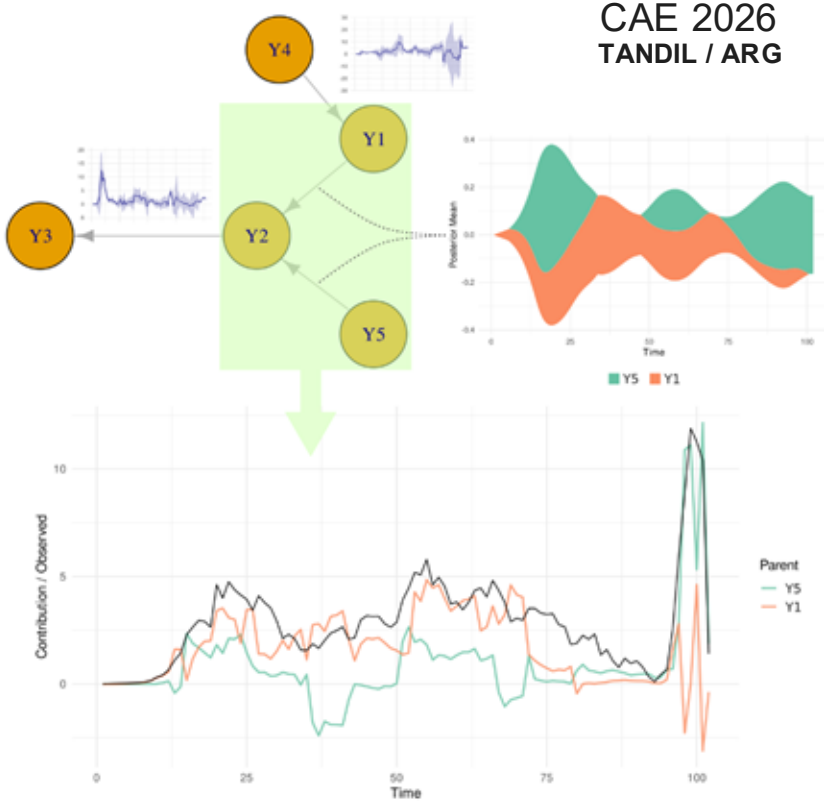
ABSTRACT

The Multiregression Dynamic Model (MDM) is a framework that combines graph theory with dynamic linear models, allowing a non-Gaussian multivariate structure to emerge in the context of causal time series. Since an optimal DAG structure is an NP-hard task, this package overcomes the all-combinations search (Integer Programming Algorithm) using heuristic algorithms (like Hill Climbing). Written using R S4 Object programming, it performs learning functions (estimating network structure and its dynamic arcs), as well as includes DAG (causal) visualization, time-varying coefficients visualization, and graphical performance checks. The MDM R package is distributed under the GPL license and is accessible from GitHub.



LEARNING

1. STRUCTURE ESTIMATION
2. DISCOUNT FACTORS SCORES
3. DLM ADJUST (filter & smooter)



Multiregression Dynamic Model with Bayes Factor (MDM-BF)

(PART 1) - RELATIONAL ESTIMATION (NETWORK STRUCTURE in local optimum –partial DAG– generated by a heuristic):

- Estimation as a **Kalman filter** [r = number of sensors]

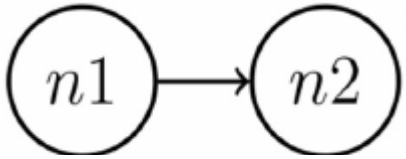
$$Y_t(r) \sim \mathcal{N}(\mathbf{F}_t(r)'\boldsymbol{\theta}_t(r), V_t(r))$$
$$\boldsymbol{\theta}_t \sim \mathcal{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t), \longrightarrow \mathbf{W}_t = \frac{1-\delta}{\delta} \times \mathbf{C}_{t-1},$$

$$\mathbb{P}(\tau, \mathbb{Y}|\mu, \sigma^2\Sigma, \nu) \propto \mathbb{P}(\mathbb{Y}|\tau, \mu, \sigma^2\Sigma)\mathbb{P}(\tau|\nu)$$
$$\mathbb{Y}|\tau, \mu, \sigma^2\Sigma \sim N(\mu, (\sigma^2\Sigma/\tau))$$
$$\tau|\nu \sim Ga(\nu/2, \nu/2)$$

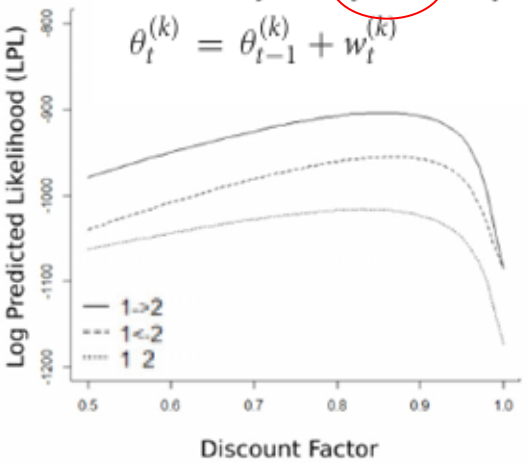
$$f(\mathbb{Y}|\mu, \sigma^2\Sigma, \nu) = \frac{\Gamma[(\nu+r)/2]}{(\pi\nu)^{r/2}|\sigma^2\Sigma|^{1/2}\Gamma[\nu/2]} \left(1 + \frac{(\mathbb{Y}-\mu)'\Sigma^{-1}(\mathbb{Y}-\mu)}{\sigma^2\nu}\right)^{-(\nu+r)/2}$$

MODEL SELECTION via Log-Predictive Likelihood (LPL) equivalente to Bayes Factor.

e.g. SIMULATION



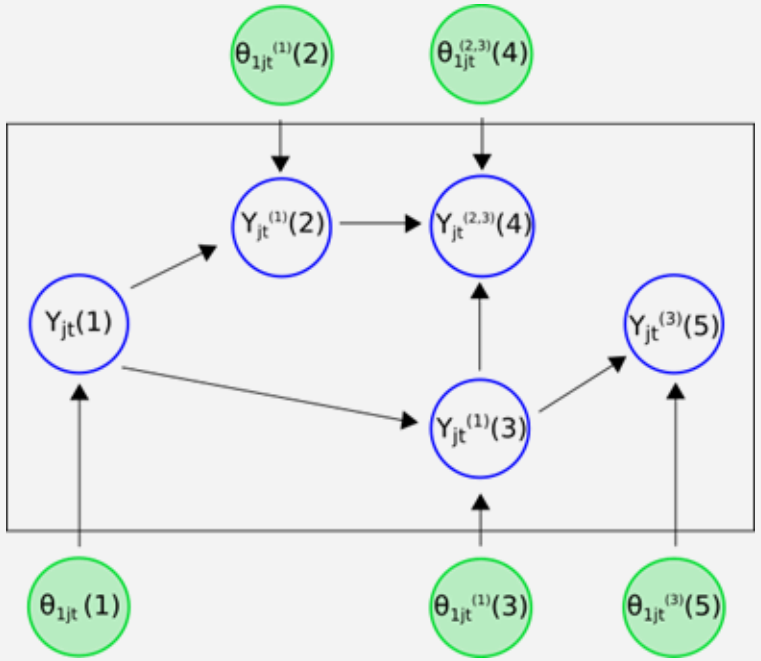
$$n1_t = \theta_t^{(1)} + v_t^{(1)}$$
$$n2_t = \theta_t^{(2)} + \theta_t^{(3)} n1_t + v_t^{(2)}$$
$$\theta_t^{(k)} = \theta_{t-1}^{(k)} + w_t^{(k)}$$



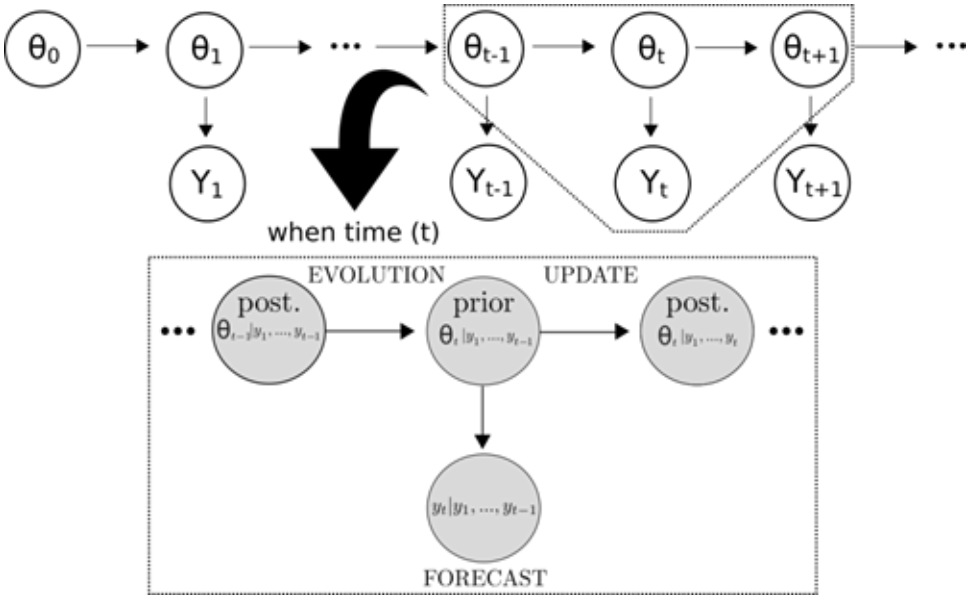
(PART 2) - TIME-VARYING ESTIMATION:

BAYESIAN DYNAMIC ESTIMATION APPROACH

PROCESS MODEL



P(Y(1),Y(2),Y(3),Y(4),Y(5)**) = P(**Y(1)**)** x **P(**Y(2)|Y(1)**)**
P(Y(3)|Y(1)**)** x **P(**Y(4)|Y(2),Y(3)**)** x **P(**Y(5)|Y(3)**)**



Group structure methods using hierarchical dynamic in Time Series

Then, NETWORK STRUCTURE + DYNAMIC STRENGTH, can be seen as a decomposition of dissimilarity matrix, d, to Embedding projection reduction space

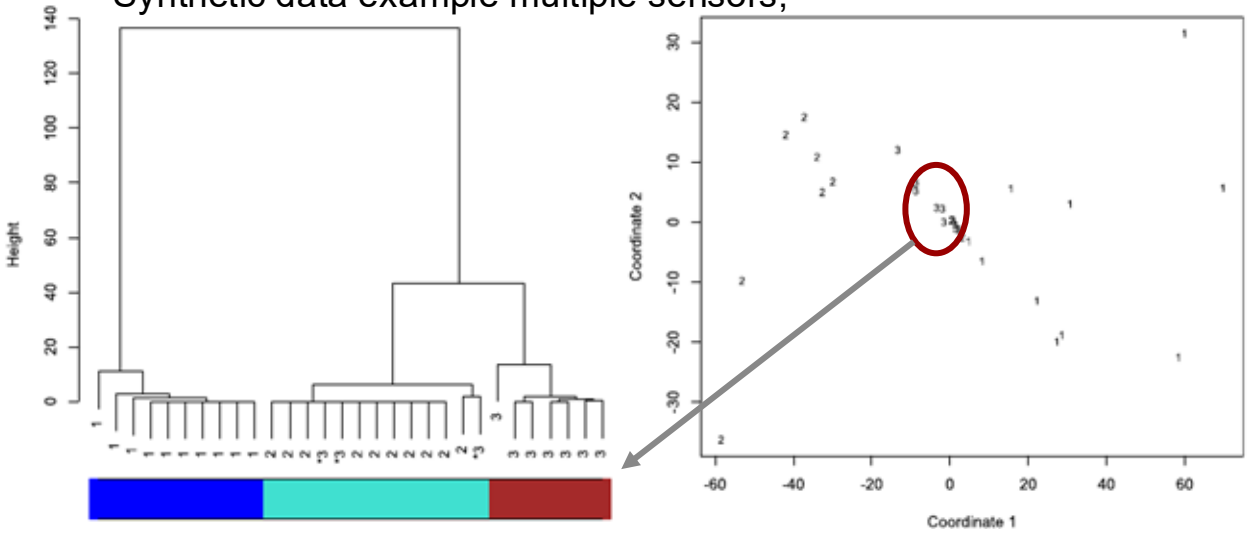
$$d(i,j) = \mathrm{LPL}_i(M_i) + \mathrm{LPL}_j(M_j) - \mathrm{LPL}_i(m_{ij}) - \mathrm{LPL}_j(m_{ij}),$$

Using this method after estimating different dependence levels

- Individual networks (IS), Mi and Mj, and
- Pairwise group network (GS), mij, for every pair of subjects i and j,

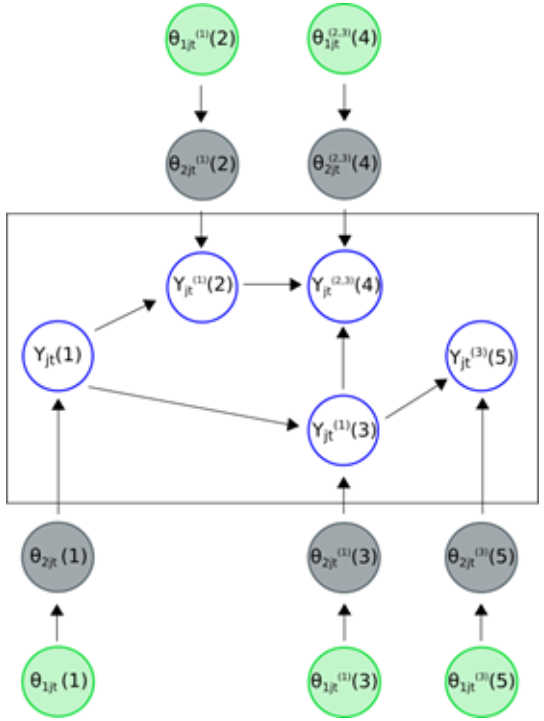
a separation measure, d(i, j).

Synthetic data example multiple sensors,



E.g. a Group Hierarchical Dynamic Model for 3 levels [machine, sensors and periods], global dynamic (GS) with individual patterns (contributions - VTS + CS + IS), can be seen as a fix Network causal structure and a dynamic strength:

- Common-structure (CS) -> Entire Process
- Virtual-typical-subject (VTS) -> Group of Machines
- Group-structure (GS) -> Machine
- Individual-structure (IS) -> Sensors



SENSORS

TIME WINDOW

DAG structure

$$\mathrm{LPL}_i(m) = \sum_{r=1}^{n} \sum_{t=1}^{T} \log p_{tr}(y_{it}(r) | \mathbf{y}_i^{t-1}, Pa_i(r), m),$$

[PARENT]

*the Network structure is estimated as a SCORE (joint Log Predictive Likelihood, LPL).



REFERENCES

CAE 2026
TANDIL / ARG

Montgomery, D. C., & Jennings, C. L. (1991). **Introduction to statistical quality control** (Vol. 5). New York: Wiley.

Chakraborti, S., & Graham, M. (2019). **Nonparametric statistical process control**. John Wiley & Sons.

Nelsen, R. B. (2006). **An introduction to Copulas**. New York, NY: Springer New York.

Bock, H. H., & Diday, E. (Eds.). (2012). **Analysis of symbolic data: exploratory methods for extracting statistical information from complex data**. Springer Science & Business Media.

West, M., & Harrison, J. (1997). **Bayesian forecasting and dynamic models**. New York, NY: Springer New York.

Queen, C.M. and Smith, J.Q. (1993). Multiregression dynamic models. **Journal of the Royal Statistical Society: Series B** (Methodological), 55(4), p.849-870.

Diego Nascimento
diego.nascimento@neoma-bs.fr

LinkedIn

THANK YOU

Especial thanks to:

NEOMA BUSINESS SCHOOL

SOCIEDAD ARGENTINA DE ESTADÍSTICA

Grupo Argentino de Bioestadística